to be switched on and paired with a smartphone having the AgroNxt Bhu-Parikshak app via Bluetooth connection. Before use, the device needs to be calibrated and 5-8 grams of soil sample is poured into the designated sample cup for testing. The results are calculated and displayed in the application, which can then be shared with the farmers—all within two minutes.

Vardhan informs, “The device is based on the near-infrared spectroscopy, which is based on molecular overtones and combination vibrations that originate from the fundamental vibrational bands generally found in the mid-IR region.” It estimates the available forms of nitrogen, phosphorus, and potassium along with the clay content, cation exchange capacity, and organic carbon in the soil. The company claims that the device can test 50,000 soil samples in its lifetime, with 100-250 samples in a single day.

“As per IIT Kanpur studies, once calibrated, the device gives results with 80-96% accuracy, which is a highly acceptable figure in the soil testing market,” according to Vardhan. He adds that upon regular usage, the results keep refining because “this machine learns quickly.”

AgroNxt CEO Rajat Vardhan showcasing Bhu-Parikshak at ASEAN-India startup Festival 2022

When the product was transferred to the startup as a minimum viable product, they took feedback from the industry partners—which included sugar mills, agricultural input companies, FMCG companies, a few farmers, and FPOs—to ensure that it was acceptable in the market.

The final hardware costs around ₹100,000, inclusive of all taxes. A nominal recurring charge of ₹10 per soil sample sustains the huge cloud infrastructure set up for this purpose.

Vardhan says that they are tying up with established players from agri-business to reach more farmers. For example, ITC is planning to work with thousands of FPOs across the country, and AgroNxt is looking forward to closely partner with the conglomerate and “support a large base of farmers in a much quicker succession.” The startup has recently tied up with various state agricultural universities to provide farmers with prompt soil testing and customised fertilizer recommendations at scale.

Interestingly, AgroNxt is looking forward to doubling its team in the next six months. “We’d like to scale Bhu-Parikshak as a global brand for soil testing and will add more features to the product from time to time,” says Vardhan. In the upcoming year, AgroNxt plans to unveil at least two more deep-tech solutions pertaining to the agriculture field (quite literally).

—*Vaishali Yadav*

2 MATRI: MAINSTREAMING MENSTRUATION WITH PAIN RELIEF

Two Bengali schoolboys used to make electronic gadgets for fun. One day in college these engineering students brainstormed about finding a solution to a common daily life problem. And they came up with Matri, a wearable electronic device to block menstrual pain—kudos, they thought about women’s problem!

In that part of the world, where even talking about menstruation is considered dirty and shameful, two innovators from Bengal, Rohan Roy and Roni Mondal, collaborated to relieve the pain of menstruating women with their unique device—Matri. But why?

Roy says, “We see our mothers and sisters facing the same problem. There are only two conventional remedies. Hot water bags, which can’t be carried in public places, and medicines have side effects. So, we brought Matri to this big and untouched market. It is a tiny, portable, rechargeable, and wearable electronic device which is completely non-invasive in nature.”

In Bengali, *Matri* means mother. “In your tough days, it takes care just like a mother, hence the name,” remarks Mondal.

Though their startup Silifarm Technologies was registered in